

Article

Integrated Control of Hybrid Thermochemical–PCM Storage for Renewable Heating and Cooling Systems in a Smart House

Georgios Martinopoulos ^{1,2,*}, Paschalis A. Gkaidatzis ³, Luis Jimeno ⁴, Alberto Belda González ⁴, Panteleimon Bakalis ⁵, George Meramveliotakis ⁵, Apostolos Gkountas ⁵, Nikolaos Tarsounas ³, Dimosthenis Ioannidis ³, Dimitrios Tzovaras ³ and Nikolaos Nikolopoulos ¹

¹ Chemical Process and Energy Resources Institute, Centre for Research & Technology Hellas, 6th km Charilaou-Thermis, GR 57001 Thermi, Greece; n.nikolopoulos@certh.gr

² Merchant Marine Academy of Macedonia, GR 57004 Néa Michanióna, Greece

³ Information Technology Institute, Centre for Research & Technology Hellas, 6th km Charilaou-Thermis, GR 57001 Thermi, Greece

⁴ Energy Division, Cartif Technology Centre, ES 47151 Valladolid, Spain; albbel@cartif.es (A.B.G.)

⁵ Psycrotherm, GR 18540 Piraeus, Greece; pbakalis@psycrotherm.gr (P.B.); gmeramveliotakis@thermodraft.gr (G.M.)

* Correspondence: g.martinopoulos@certh.gr

Abstract

The development of integrated renewable energy and high-density thermal energy storage systems has been fueled by the need for environmentally friendly heating and cooling in buildings. In this paper, MiniStor, a hybrid thermochemical and phase-change material storage system, is presented. It is equipped with a heat pump, advanced electronics-enabled control, photovoltaic–thermal panels, and flat-plate solar collectors. To optimize energy flows, regulate charging and discharging cycles, and maintain operational stability under fluctuating solar irradiance and building loads, the system utilizes state-of-the-art power electronics, variable-frequency drives and modular multi-level converters. The hybrid storage is safely, reliably, and efficiently integrated with building HVAC requirements owing to a multi-layer control architecture that is implemented via Internet of Things and SCADA platforms that allow for real-time monitoring, predictive operation, and fault detection. Data from the MiniStor prototype demonstrate effective thermal–electrical coordination, controlled energy consumption, and high responsiveness to dynamic environmental and demand conditions. The findings highlight the vital role that digital control, modern electronics, and Internet of Things-enabled supervision play in connecting small, high-density thermal storage and renewable energy generation. This strategy demonstrates the promise of electronics-driven integration for next-generation renewable energy solutions and provides a scalable route toward intelligent, robust, and effective building energy systems.

Keywords: thermal energy storage; thermochemical storage; building energy management system; control algorithms; iot-enabled energy management; renewable energy integration

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1. Introduction

Around 40% of the European Union’s final energy demand is used for heating and cooling in buildings [1], with a growing focus on integrating renewable energy sources

(RES) to meet this demand. Among these, solar thermal and photovoltaic systems are the most widely used technologies [2], offering a competitive alternative to fossil fuels—especially in southern Europe, where solar irradiance is high [3]. However, the intermittent nature of solar energy necessitates the use of thermal energy storage (TES) systems to ensure year-round supply [4]. TES technologies are categorized into sensible heat, latent heat, and thermochemical (TCM) storage. Sensible heat systems are common but have low energy density (20–30 kWh/m³), requiring large volumes [5,6]. Latent heat systems using phase-change materials (PCMs) offer higher energy density (40–80 kWh/m³) but are still bulkier compared to TCM systems [7]. TCM storage stands out with energy densities between 100 and 600 kWh/m³, making it suitable for compact, long-term storage in residential settings [8]. Advantages include low heat losses and seasonal storage potential, but challenges such as high costs, slow reaction kinetics, material degradation, and system complexity remain [6,9]. Research on TCM storage is expanding, with both experimental and numerical studies exploring materials like CaCl₂–NH₃ [10–12], though most are limited to laboratory-scale prototypes. Numerical simulations have modeled TCM kinetics and integrated storage systems [13,14]. Center for Research and Technology Hellas (CERTH) has developed a dynamic model of a compact TCM–PCM hybrid storage system called MiniStor, using Aspen Plus Dynamics and Matlab/Simulink [10]. As part of the Horizon 2020 project MiniStor [15], a working prototype has been deployed at CERTH’s Smart Home in Thessaloniki [16].

Advances in power electronics have become critical enablers for optimizing the performance, control, and reliability of such renewable energy systems and their associated TES solutions. In particular, power electronic components such as wide bandgap semiconductors, for example, silicon carbide (SiC) and gallium nitride (GaN), offer higher efficiency, higher switching frequencies, and improved thermal management, enabling more compact and reliable converters for integrating PV, solar thermal, and TES technologies into building energy systems.

Innovative circuit topologies such as multi-level converters and soft-switching designs minimize switching losses and electromagnetic interference, improving system efficiency and lifespan, which is crucial for residential and commercial heating and cooling applications. The modular design of power electronic systems facilitates scalability and maintenance, enhancing uptime and adaptability in smart grid environments.

Control strategies increasingly leverage predictive and adaptive algorithms to manage energy flow dynamically, optimizing charging and discharging cycles of TES while coordinating with the electrical grid or smart building energy management systems. Advanced digital control processors, embedded sensors, and IoT connectivity support real-time monitoring, fault detection, and predictive maintenance, enhancing reliability and longevity of both power electronics and TES units.

Energy storage control systems have been widely studied by energy practitioners. PCMs have been widely studied for latent thermal energy storage, but control strategy research is focused primarily on building and heating, ventilation, and air conditioning integration rather than hybrid TES systems. In PCMs, control strategies usually aim to optimize charging/discharging to reduce energy consumption and peak demand. They are either rule-based (on temperature thresholds or time schedules), optimization-based (through predictive control within building energy management systems), or demand-response-based [17].

With TCM, on the other hand, due to its complexity, as it relies on reversible chemical reactions for storing and releasing energy, explicit control systems are less mature and often rely on the control of the chemical reaction rather than a dedicated hybrid TES framework [18].

The current literature shows a clear interest in PCM control strategies and a recognized potential for advanced control in TCM applications, but hybrid TCM and PCM systems remain at an early stage of dedicated control system research [19] and this is the gap that this paper fills with the results from the operation of the MiniStor System.

The transition toward climate-neutral and resilient energy systems requires the seamless integration of renewable energy sources, advanced thermal energy storage technologies, and intelligent power electronics within unified and controllable architectures. In this context, the Horizon Europe MINISTOR project developed and demonstrated an innovative solution for the efficient coupling of various energy systems, enabling increased flexibility, reliability, and sustainability across residential and small-scale applications.

In this paper, the integration of advanced power electronics into renewable energy and thermal storage systems is presented as implemented in the MiniStor prototype installed at CERTH's Smart Home in Thessaloniki, Greece. This integration, combined with advanced modeling, control, and reliability techniques, accelerates the transition toward more sustainable, efficient, and resilient energy solutions across diverse sectors. The MiniStor prototype successfully integrates flat-plate solar collectors, photovoltaic–thermal (PVT) panels, and a hybrid TCM–PCM thermal storage system with intelligent power electronics and control, demonstrating tangible progress toward future smart energy systems. The paper presents the system components, the monitoring and control architecture, and an overview of the initial operational results. The main contributions of this work are summarized as follows:

- **Hybrid Thermal Energy Storage Innovation.** The hybrid TCM–PCM thermal storage concept is validated under real-life operating conditions. By combining thermochemical materials, offering high energy density and long-term storage capability, with phase-change materials, providing high power density and short-term buffering, key limitations of conventional thermal energy storage technologies are addressed, enabling flexible and efficient storage across multiple time scales.
- **Integrated Multi-Energy System Architecture.** The paper presents a holistic integration of electrical and thermal energy domains by coupling solar thermal collectors, photovoltaic–thermal (PVT) systems, hybrid thermal storage, and advanced power electronics within a unified system architecture. This approach supports sector coupling and coordinated multi-energy operation.
- **Advanced Power Electronics and Control.** Intelligent power electronic interfaces and hierarchical control strategies are presented, enabling optimal energy flow management between electrical and thermal subsystems, as well as improved reliability assessment and fault awareness.
- **Real-World Demonstration and Validation.** A key contribution of this work is the real-world validation of the proposed proof of concept. The experimental results confirm the technical feasibility of the hybrid storage approach and demonstrate its operational benefits in realistic environments.

This paper is organized as follows: In Section 2, the description of the system along with its control, both in hardware and software, and monitoring systems are presented. In Section 3, the data management approach adopted for this system is described. In Section 4, the system during operation is analyzed. Lastly, Section 5 is devoted to the conclusions.

2. Materials and Methods

2.1. Description of the System

The MiniStor system is composed of four main subsystems:

1. Solar Field—includes photovoltaic-thermal (PVT) and flat-plate solar (FPC) thermal collectors, along with a buffer tank for heat generation.
2. TCM Reactor—a thermochemical reactor paired with an ammonia (NH_3) sorption cycle.
3. Heat Pump and PCM Storage—a heat pump integrated with both hot and cold phase-change material (PCM) tanks.
4. Battery System—powered by electricity from the PVTs via an inverter to cover internal electrical loads (e.g., pumps, compressors).

An overview of the subsystems and their integration with the building's heating, ventilation, and air conditioning (HVAC) system is shown in Figure 1 [10].

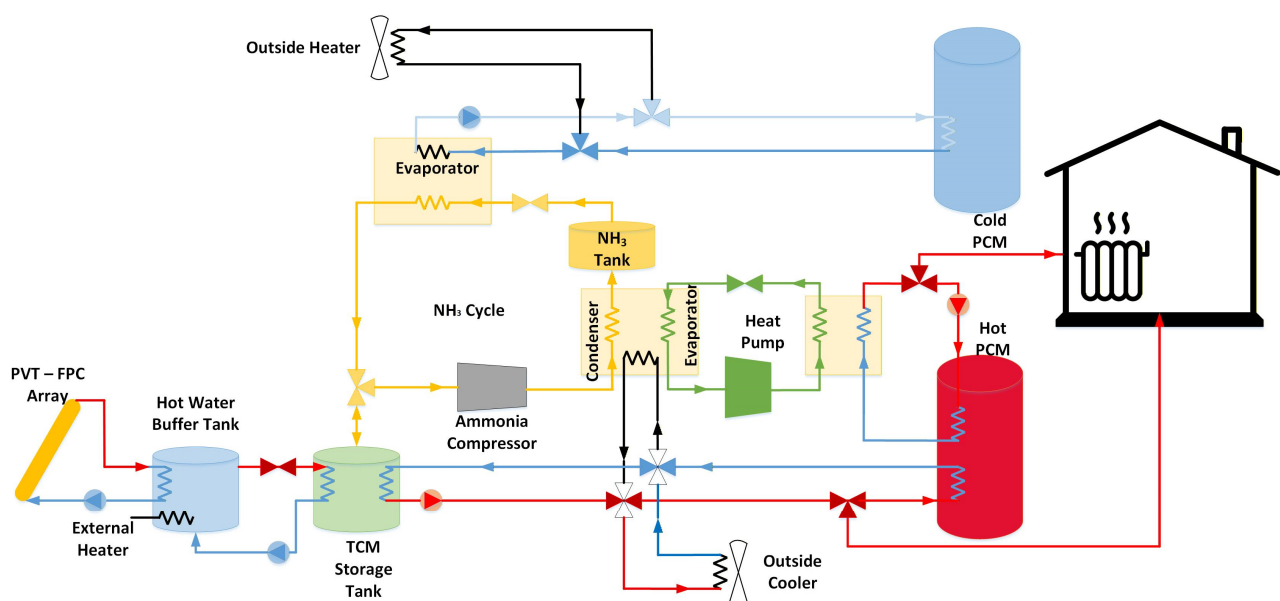


Figure 1. Overview of MiniStor thermal system components.

MiniStor is installed at CErTH's Smart Home in Thessaloniki, Greece (40.56° N, 22.99° E). The building has two floors and 317.7 m² of habitable area (182.7 m² ground floor, 135 m² first floor). Currently used as an office space, it operates mainly during business hours. MiniStor covers the heating and cooling requirements of a 49 m² room via a dedicated fan coil unit (6 kW heating, 4 kW cooling) directly connected to the PCM tanks. All PCM tanks are filled with materials based on sodium acetate trihydrate. Piping between the solar field, container, and Smart Home is insulated and buried to minimize heat losses.

In winter, heat is supplied in two modes:

Charging Mode: Heat for salt desorption is primarily provided by the solar field, with a 2 kW_e auxiliary heater if needed. Heat input initiates the endothermic decomposition reactions in the TCM reactor, generating gaseous NH_3 . The gas is compressed, condensed, and stored. The condensation heat is upgraded by a heat pump to 336 K to meet building heating demands.

Discharging Mode: This mode is used during nights or cloudy days. Stored NH_3 evaporates when ambient temperature is lower than the reactor's equilibrium pressure. The resulting exothermic reaction releases heat (330–336 K) in the reactor, which is stored in the hot PCM tank.

During summer, part of the cooling load is met via NH_3 evaporation during the discharging phase. The cold output is stored in a PCM tank with a melting temperature of 284 K.

All components are housed in a 17.3 m³ container equipped with comprehensive safety systems (Figure 2). The solar array consists of the following equipment:

- 10 PVTs: each with 1.55 m² aperture area, optical efficiency 0.51, heat loss coefficients $\alpha_1 = 4.93 \text{ W/m}^2\text{K}$ and $\alpha_2 = 0.021 \text{ W/m}^2\text{K}$, and 260 W peak power;
- 5 FPCs: each with 2.37 m² gross area, optical efficiency 0.823, and heat loss coefficients $\alpha_1 = 3.36 \text{ W/m}^2\text{K}$ and $\alpha_2 = 0.013 \text{ W/m}^2\text{K}$.

The PVTs and FPCs are installed with a 37° inclination, facing south (Figure 2). An external heat exchanger prevents overheating of the PVTs during summer. A water buffer tank and a 2 kW_e heater support stable thermal input. An automation system manages thermal fluid recirculation based on solar output, ensuring that the TCM storage receives consistent heat. The TCM reactor, a key MiniStor innovation, is based on a reversible reaction between NH_3 and CaCl_2 .

The reactor contains seven 1.25 m long tubes (114.3 mm in diameter), forming two subreactors with a combined thermal capacity of 17.5 kWh. $\text{CaCl}_2 \cdot 4\text{NH}_3$ is used as it offers superior heat storage performance compared to zeolites [20].



Figure 2. Solar array field comprising 10 PVTs and 5 FPCs.

A custom R134a heat pump boosts NH_3 condenser heat from 301 K to 336 K for use in the HVAC or storage in the hot PCM. The refrigerant cycle involves compression (to 4200 kPa at 294 K), condensation (at 337 K), and expansion. The cooling/heating demand of the building is covered via a dedicated fan coil which absorbs heat/cold from the two PCM tanks: a hot PCM tank that stores 3.5 kWh at 331 K (volume: 0.09 m³) and supplies the fan coil via insulated PPA piping and a cold PCM tank that stores 6 kWh at 284 K (volume: 0.22 m³), absorbing surplus cooling from NH_3 evaporation. MiniStor complies with EN 378 [21], ensuring NH_3 safety (below 50 kg charge). All NH_3 equipment is housed in a separate, restricted compartment with emergency access, lighting, and alarms (Figure 3).



Figure 3. MiniStor NH₃ container.

2.2. Control of the System (Hardware)

The thermochemical subsystem uses NH₃ (R717) as the working fluid and includes the TCM tank, an NH₃ receiver, an oil separator, a flooded evaporator, an NH₃ separator, and multiple valves and sensors, as shown in Figure 4. During the charging mode, NH₃ vapor is extracted from the TCM tank by a variable-speed compressor and condensed into the NH₃ receiver. During discharging, liquid NH₃ is expanded through an EEV (electronic expansion valve) into a flooded evaporator connected to the cold PCM tank. Critical auxiliary hardware such as a desuperheating valve, a hot gas bypass line, and an oil-return solenoid ensure stable operation under open-loop dynamics. Key measurements include pressure, level, and temperature sensors throughout the NH₃ circuit.

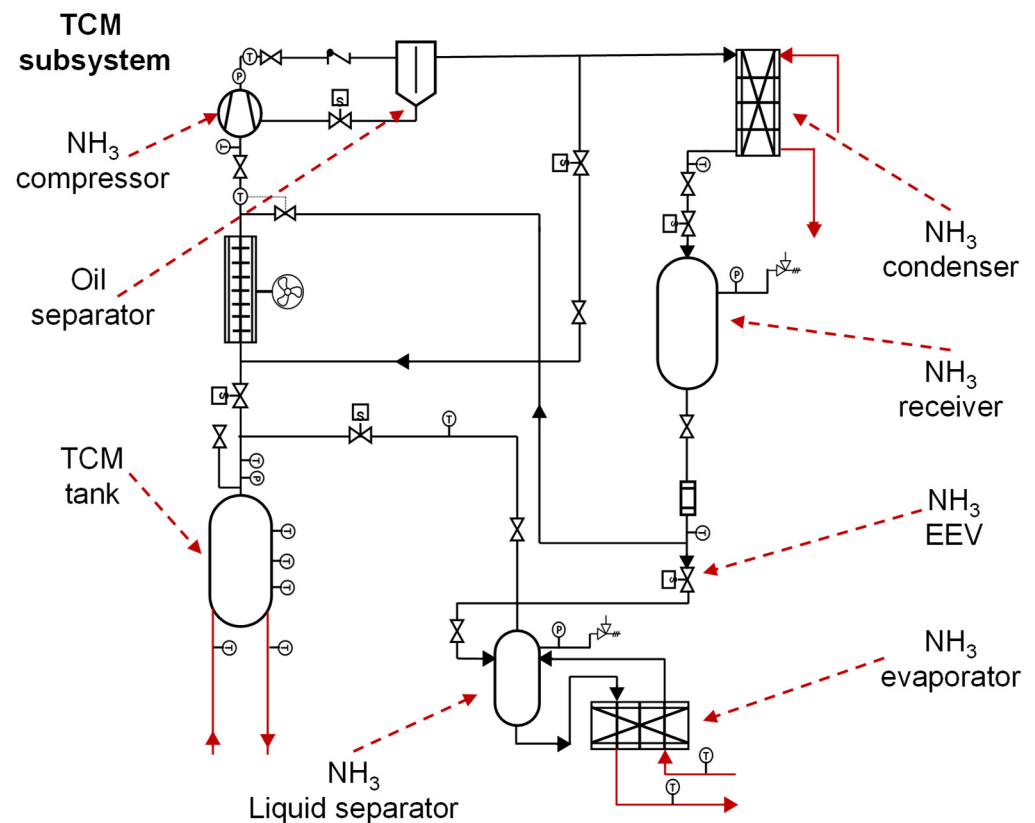


Figure 4. Schematic diagram and main components of TCM subsystem.

The heat pump subsystem, as shown in Figure 5, is based on a compact R134a loop. It receives thermal energy either from the NH₃ condenser or the HTF (heat transfer fluid) buffer and upgrades it to deliver hot water at temperatures above 333 K. The condenser of the NH₃ circuit is coupled thermally to the heat pump evaporator, enabling high-grade heat recovery.

The third subsystem, the HTF circuit, uses a water–propylene glycol mixture (3 to 1 ratio) and serves as a common heat transfer medium between the solar array, TCM tank, heat pump, and thermal PCM buffers. It is driven by multiple circulation pumps and features control valves, strainers, and expansion vessels. The hydraulic topology allows dynamic routing of heat to or from the TCM unit, hot PCM, DHW tank, or cold PCM modules.

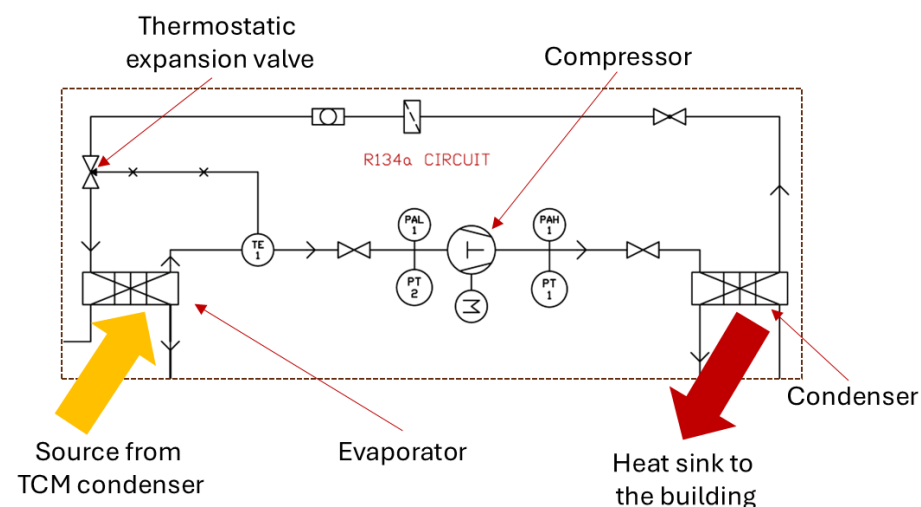


Figure 5. Schematic diagram and main components of the heat pump subsystem.

To enable precise monitoring and control of the MiniStor subsystems, an extensive instrumentation layer is implemented across the entire system (Figures 4 and 5). The NH₃ circuit includes multiple pressure transmitters, temperature sensors, and level switches strategically placed at the TCM tank, NH₃ separator, receiver, and compressor inlet/outlet. The key TCM sensors together with their measurement point, range, type, and accuracy are listed in Table 1.

Table 1. List of sensors used in the TCM subsystem.

Subsystem Point	Measured Variable	Sensor Type	Nominal Range	Accuracy
Pressure transmitter—high	NH ₃ pressure (high side)	Danfoss AKS 32R pressure transmitter, 4–20 mA	0–400 PSI	±0.3% FS
Pressure transmitter—low	NH ₃ pressure (low side)	Danfoss AKS 32R pressure transmitter, 4–20 mA	0–200 PSI	±0.3% FS
Temperature sensor—NH ₃	NH ₃ lines	NTC, 2-wire	223...423 K	±0.1 K
NH ₃ level transmitter	Receiver liquid level	NIVOCAP CT-200, 4–20 mA, 2-wire	0–100%	±0.3%
Temperature sensor—water	HTF/Water (reactor jacket/HX)	Pt100 RTD, 3-wire	253...393 K	±0.1 K

These sensors allow real-time tracking of pressure ratio, suction/discharge temperatures, and refrigerant charge condition. Flow switches and temperature sensors are also deployed in the HTF circuit to confirm flow rate control and validate thermal gradients

across components. Safety instrumentation includes an NH_3 detector, pressure safety valves, high-pressure cut-offs, and level-based interlocks to prevent overfilling or compressor dry-run conditions. All analogue and digital sensor inputs are processed by the slave PLC (low-level communication), as shown in Figure 6, which also communicates with a higher-level SmartServer IoT PLC. Communication is performed via Modbus TCP. This layered instrumentation approach ensures that all operational modes are executed within the safe and efficient operating window, while also enabling performance monitoring and fault diagnostics.

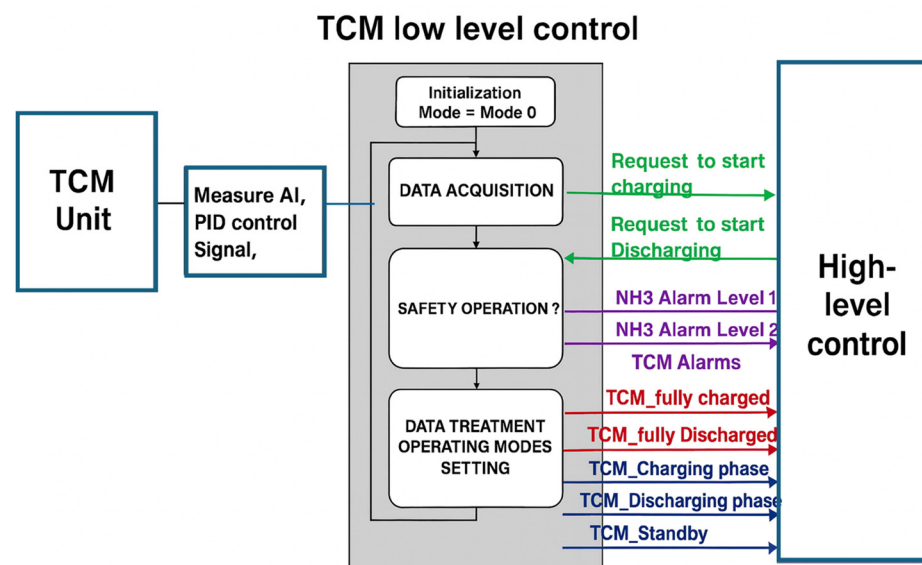


Figure 6. Control flowchart between low- and high-level communication of the TCM.

The SmartServer IoT programmable logic controller (PLC) serves as the main controller of the MiniStor system, responsible for local control logic and managing communication between sensors, actuators, and external monitoring platforms. This controller communicates via the Modbus TCP/IP protocol with equipment related to the solar field, such as the hybrid inverter and the specific controller for that part of the system. Communication with these devices is unidirectional in terms of control, as it is limited to data reading for monitoring purposes, without directly acting upon them.

In parallel, the system includes a second controller, a Mitsubishi slave PLC, which handles the lowest-level system operations. This controller's main function is to process signals directly from physical sensors and actuators.

In addition to its control functions, the SmartServer IoT consolidates all acquired signals, both from sensors and other connected devices, and retransmits them using different protocols. On the one hand, it sends data to a technical visualization platform based on Grafana via the message queuing telemetry transport (MQTT) protocol. On the other hand, the SmartServer IoT sends selected data via HTTP requests through a representational state transfer application programming interface (REST API) to a second platform, which is designed for simplified visualization by the user.

Hardware Subsystems and Auxiliary Control Mechanisms of the TCM unit: In addition to the main thermochemical energy storage operation, several hardware subsystems are integrated into the TCM- NH_3 circuit to ensure safe and stable operation of the compressor (Figure 7). These subsystems address the unique requirements arising from the open-loop nature of the circuit and are actively managed by the TCM controller based on real-time process conditions such as compressor runtime, pressure ratio, and discharge temperature.

Oil Management and Recovery: As the NH_3 circuit operates in open-loop mode—unlike the typical closed-loop refrigeration cycles—oil return to the compressor cannot rely on standard internal circulation. To address this, an oil separator is installed on the compressor discharge line. An oil solenoid valve is used to control the return of separated oil to the compressor. This valve is periodically triggered by the TCM controller, depending on the compressor operating time and the calculated pressure ratio, ensuring continuous lubrication and compressor protection. With the compressor running, when the discharge to suction pressure ratio (PR) exceeds the configured setpoint, the oil-return solenoid valve is activated for 2 sec and then is inhibited for 60 sec before any subsequent activation. In addition, if PR stays above the setpoint continuously for 60 sec, a single pulse opens the valve for 2 sec (high pressure ratio assist). The valve's actual state is fed back to the trigger logic to block overlapping commands and prevent immediate retriggers, ensuring clean, time-accurate pulses.

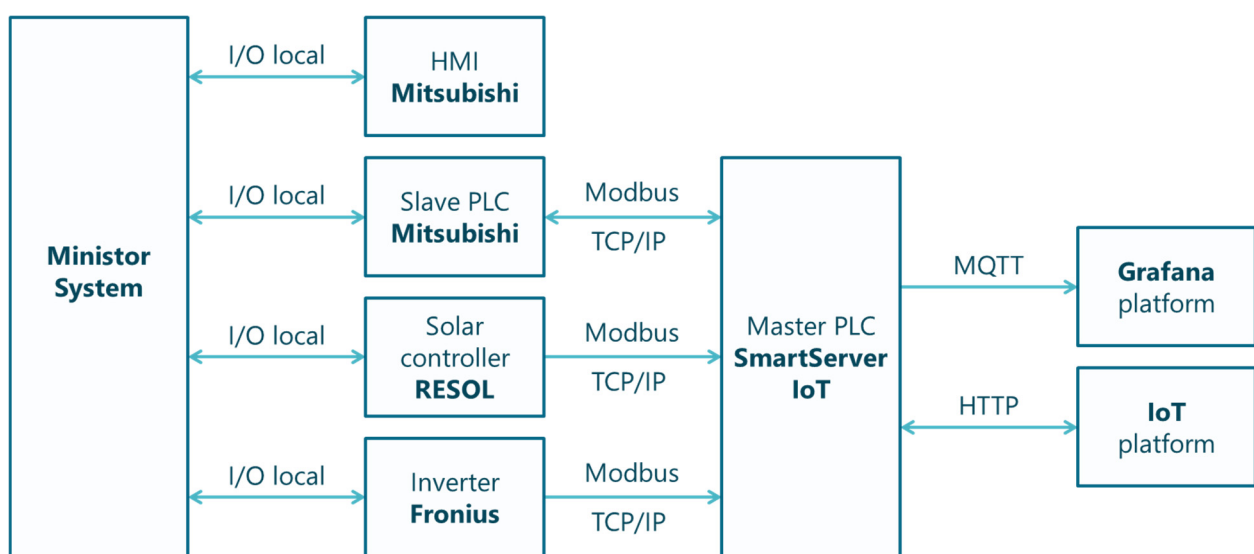


Figure 7. Hardware configuration.

Compressor Pressure Ratio Control: The compressor's rotational speed is regulated using a variable frequency drive (VFD) in order to maintain the desired pressure ratio during the charging mode. If VFD regulation alone is insufficient to keep the pressure ratio below the defined limit (typically $\text{PR} < 5.5$), a hot gas bypass valve is activated. This valve redirects a portion of the compressor's discharge vapor back to the suction line, thereby reducing the effective pressure ratio and keeping the compressor within its safe operational envelope.

Thermal Protection Measures: To prevent excessive superheat at the compressor suction, especially when high-temperature solar heat is supplied to the TCM, a desuperheating mechanism is implemented. This consists of a thermostatic expansion valve combined with a solenoid valve, installed on the compressor suction line. When the compressor discharge temperature exceeds a set threshold (typically 363 K), the solenoid is activated for 2 sec, to expand liquid NH_3 into the suction line. This evaporating liquid reduces the suction gas temperature, providing thermal protection for the compressor.

2.3. Control of the Systems (Software)

To ensure comprehensive and efficient control of the thermal system, a modular architecture was developed, composed of independent yet interconnected functional submodules, each addressing a specific system component.

One of these submodules is responsible for thermal demand management, with its primary function being to determine whether the system is capable of supplying heat to the end user. This decision is based on real-time sensor data and weather forecasts, allowing for dynamic evaluation of current and future operating conditions.

Another submodule oversees solar thermal generation, coordinating its activation based on the availability of solar resources and the system's temperature levels. This component enables efficient utilization of solar energy by adjusting its use according to production conditions.

A third submodule manages the thermal batteries (PCMs) used for both heating and cooling. Its control logic analyzes the state of charge in the storage units, the availability of thermal sources, and signals from the PCM controller to determine optimal charging and discharging times. It also processes heat discharge requests based on system demand and the operational status of associated devices.

The fourth submodule is dedicated to the control of the TCM reactor, managing its charging and discharging cycles through a rule-based logic system. This logic relies on key signals such as internal temperatures, state of charge, operational status, and energy availability. Based on these signals, the operational state of the TCM is determined and subsequently interpreted by the system's overall state machine.

All the described submodules generate status signals that are interpreted by a state machine. This state machine governs the transitions between the system's operational modes, based on logical conditions that combine inputs from the four submodules, physical sensors, and weather data. Each transition is defined as a logical expression that simultaneously evaluates multiple system variables. All this code was implemented in JavaScript using Node-RED (Figure 8).

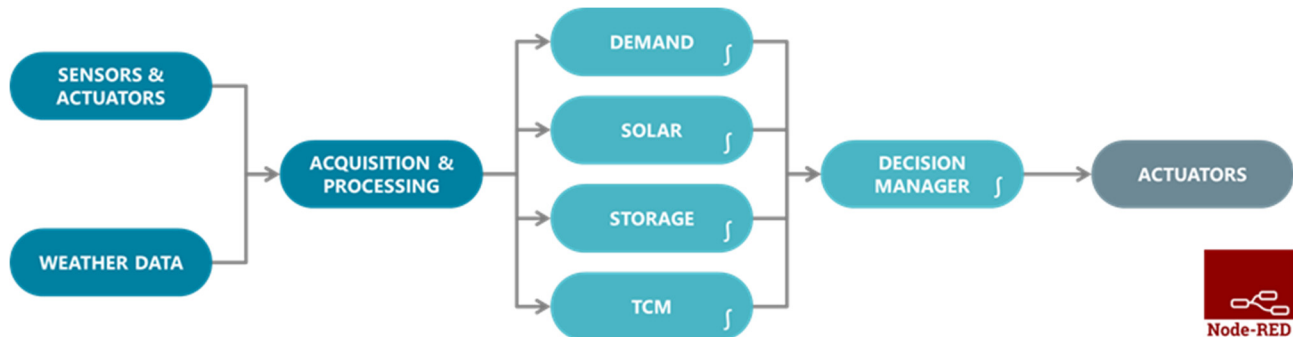


Figure 8. Energy management system (EMS) control submodules and data flow in Node-RED.

From a methodological perspective, the MiniStor EMS implements a hierarchical, rule-based control strategy organized as a finite-state machine. The upper supervisory layer encodes the admissible operating modes of the system (standby, pressurization/pre-heat, charging, depressurization, discharging, and safety modes) and the logical conditions for transitions between them, including time delays and hysteresis bands that prevent chattering. The lower layer comprises the four functional submodules and local regulators that act on individual subsystems (TCM reactor, PCM tanks, HP and solar/HTF loop). In this layer, on–off rules and threshold-based logic controllers are used to track temperature and pressure setpoints, maintain the compressor pressure ratio within its safe envelope, and enforce operational limits such as maximum condenser and suction temperatures. Together, this architecture provides a transparent mapping from measurements and user requests to actuator commands and ensures that every operating mode corresponds to a well-defined, reproducible configuration of pumps, valves, and compressors.

The adopted control approach belongs to the class of deterministic, rule-based supervisory energy management systems coordinated through a finite-state machine. Control decisions are derived from explicit logical rules formulated a priori, based on thermodynamic constraints of the $\text{CaCl}_2\text{--NH}_3$ reaction, phase-change temperature limits of the PCM units, compressor operating envelopes, and safety requirements of the NH_3 circuit. System operation is therefore governed by a discrete set of admissible operating modes and submodes, each associated with a predefined configuration of actuators and safety interlocks. This choice reflects the prototype-oriented nature of MiniStor and the need for robust, transparent, and certifiable control behavior under highly coupled thermofluid dynamics and variable solar input.

The rule set implemented at both supervisory and local levels was synthesized from a detailed analysis of system operating conditions, as documented during the design phase. In particular, temperature and pressure thresholds were defined to ensure: (i) sufficient vapor pressure for stable NH_3 circulation during charging and discharging, (ii) compliance with compressor pressure-ratio limits, (iii) compatibility between heat-source and heat-sink temperatures at the heat pump interfaces, and (iv) reliable phase-change processes within the PCM storage units. Transition rules incorporate minimum dwell times, hysteresis bands, and conditional delays to prevent oscillatory behavior caused by transient solar fluctuations or short-term load variations.

During the monitoring and commissioning phase of the MiniStor demonstrator, it became necessary to adjust low-level control strategies to ensure safe and efficient operation under real-world conditions. Although the system was properly sized, specific adaptations were required to respond to local climate variability, demand profiles, and the behavior of the installed subsystems.

The main improvements included revising control modes, redefining operational thresholds, and optimizing fault-handling logic to enhance system responsiveness and robustness. One of the most relevant changes was the implementation of a solar safety mode as a backup in case of failure of the primary controller, which acts autonomously upon detection of critical temperatures to protect thermal components.

Adjustments were also made to correct thermal imbalances affecting the heat pump and storage systems. A pre-heating mode was introduced to ensure the heat pump operates within optimal ranges before initiating the thermochemical storage charging. In addition, summer-specific dissipation strategies were added to prevent shutdowns due to overheating.

In the TCM subsystem, control logic was stabilized by enforcing minimum durations in pressurization phases, avoiding oscillatory behavior caused by solar fluctuations. Pressure and temperature limits were refined, and the NH_3 tank charge thresholds were adjusted to better reflect the system's reaction kinetics.

Finally, the charging strategy for the PCM storage was revised to eliminate thermal losses and reduce charging times. It was observed that simultaneous charging and discharging hindered phase transitions, so this condition was discouraged to improve thermal storage efficiency.

Regarding the local control of the TCM and NH_3 circuit, the slave PLC activates one of the following operational modes based on commands received from the SmartServer IoT:

Standby Mode: The TCM system remains inactive when no charging or discharging command is received from the SmartServer IoT, or if a safety function is triggered.

Pressurization Mode (Figure 9): Upon receipt of a TCM start charging command, the TCM reactor is isolated (all process valves closed) and the HTF pump circulates hot water from the solar buffer through the reactor, increasing its temperature to a bounded setpoint ($\leq 348\text{--}353\text{ K}$). The control system first evaluates the TCM tank pressure P_{TCM} . If

the tank pressure is below the predefined minimum threshold, the system enters a pressurization mode, during which the NH_3 compressor remains inactive and the pressure within the TCM tank is gradually increased until the minimum pressure level is reached. This controlled pre-heating raises the reactor vapor pressure, boosting compressor suction pressure and thereby keeping the compressor within its allowable pressure-ratio envelope. This mode ends when the tank pressure exceeds 2.5 bar and resumes if the compressor pressure ratio rises above 5.5.

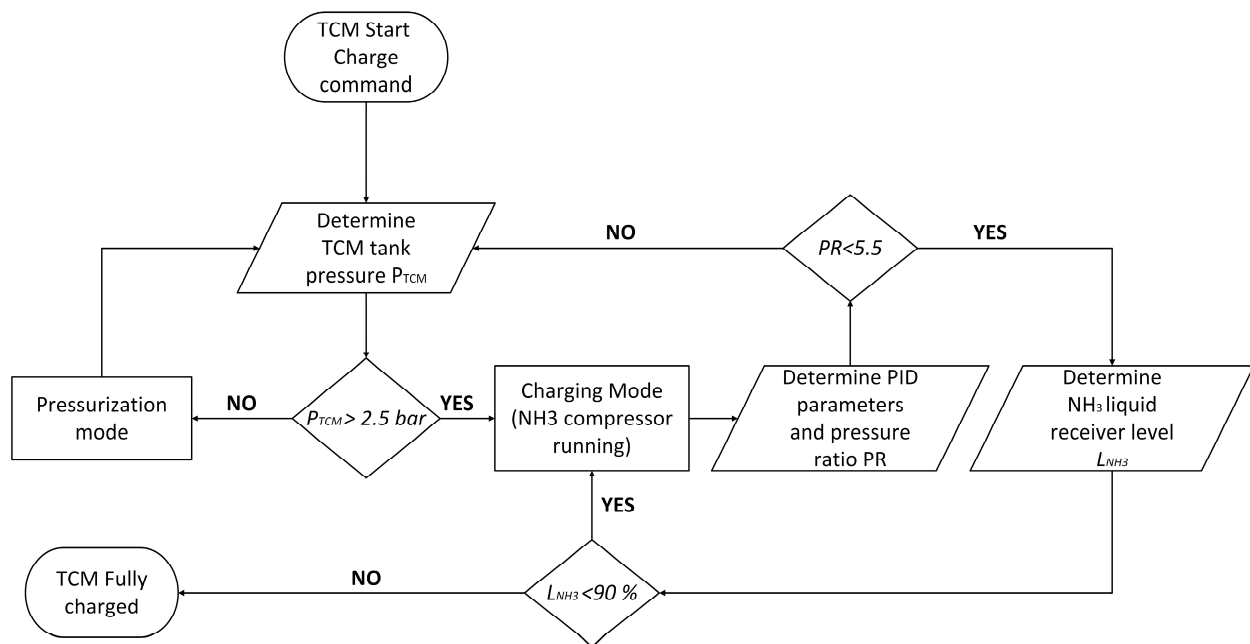


Figure 9. Flowchart of TCM charging mode logic.

Charging Mode (Figure 9): Once the condition $P_{\text{TCM}} > 2.5 \text{ bar}$ is satisfied, the system transitions to the charging mode, activating the NH_3 compressor. The compressor draws NH_3 vapor from the TCM tank, compresses it, and condenses it into liquid, which is stored in the NH_3 receiver. The TCM tank absorbs low-temperature heat from the solar buffer, while the condenser heat is upgraded by the heat pump ($>333 \text{ K}$) and stored in either the PCM for heating or the PCM for domestic hot water (DHW), based on remote control input. According to Figure 9, during this phase, the controller continuously monitors the pressure ratio (PR) across the compressor. If the pressure ratio remains below the upper safety limit ($\text{PR} < 5.5$), the controller dynamically determines the PID parameters (compressor rotational speed) and adjusts the operating pressure ratio to maintain stable performance. In parallel, the NH_3 liquid receiver level L_{NH_3} is continuously assessed. The charging process is allowed to continue as long as the receiver liquid level remains below 90%, ensuring sufficient volumetric capacity. If the condition $L_{\text{NH}_3} < 90\%$ is satisfied, the system maintains charging operation; otherwise, charging is terminated. The charging sequence concludes when the control logic determines that further charging is no longer permissible, either due to reaching the receiver level limit or completion of the intended charging process. At this point, the system transitions to a “TCM fully charged” state.

Depressurization Mode (Figure 10): The TCM tank is actively cooled to lower its pressure in preparation for discharging. At the beginning of the discharging process, the controller determines the pressure difference between the NH_3 separator and the TCM tank, defined as $\Delta P = P_{\text{NH}_3} - P_{\text{TCM}}$. This pressure differential serves as the primary criterion for enabling effective mass transfer during the discharging phase. Once the condi-

tion $P_{\text{NH}_3} - P_{\text{TCM}} > 0.2$ bar is satisfied, the control system transitions to the discharging mode.

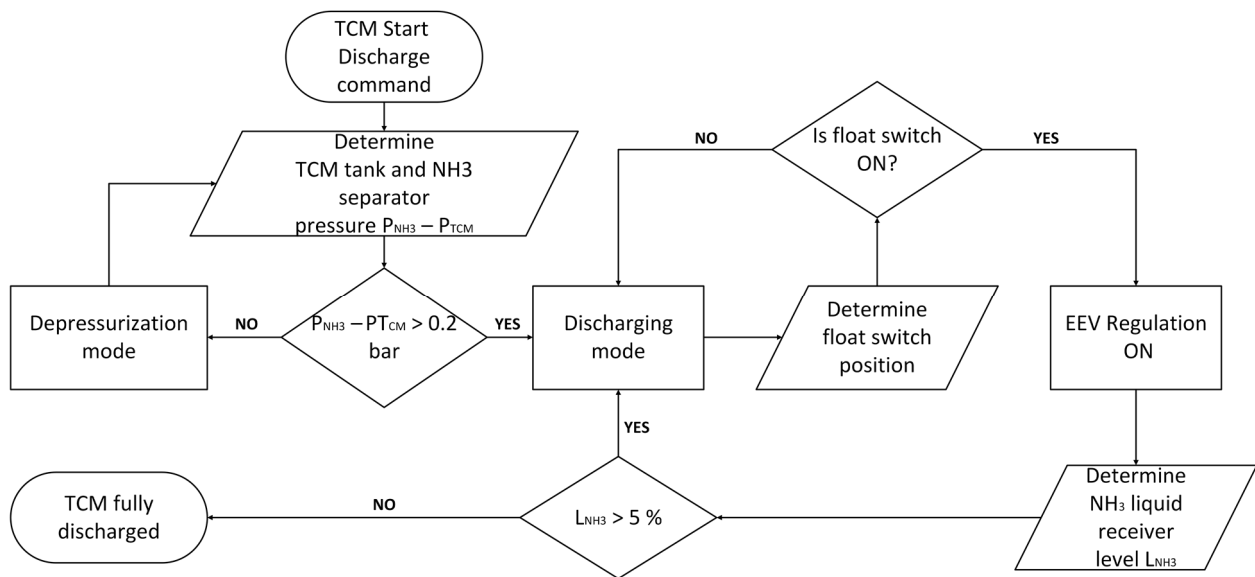


Figure 10. Flowchart of TCM discharging mode logic.

Discharging Mode (Figure 10): When the depressurization mode ends, the control system transitions to the discharging mode, allowing liquid NH_3 from the receiver to be expanded via an electronic expansion valve (EEV) into the NH_3 separator, where it evaporates using a flooded evaporator. The resulting cold from the evaporation is stored in the cold PCM tank. During this phase, according to Figure 10, the supervisory logic continuously monitors the float switch status installed in the NH_3 separator. The float switch provides a discrete indication of liquid presence and is used as a measure to avoid excessive liquid accumulation downstream. If the float switch is detected as active (ON), the controller enables EEV regulation, modulating the refrigerant mass flow rate in order to maintain stable operating conditions within the NH_3 . Conversely, if the float switch is detected as inactive (OFF), the EEV is commanded to close and no additional refrigerant is supplied from the liquid receiver. In this case, the discharging process continues by utilizing the liquid NH_3 already accumulated in the separator until the liquid level is reduced to the threshold defined by the float switch, at which point normal EEV regulation will be re-enabled. In parallel, the NH_3 liquid receiver level L_{NH_3} is continuously assessed. The discharging process is permitted to continue as long as the liquid level remains above the lower threshold of 5%, ensuring sufficient liquid refrigerant capacity for stable system operation. As long as the condition $L_{\text{NH}_3} > 5\%$ is fulfilled, the system maintains discharging mode operation. When the receiver liquid level drops below this threshold, the controller terminates the discharging sequence and transitions the system to a “TCM fully discharged” state.

To enhance usability, monitoring, and diagnostics, the MiniStor system is equipped with a dedicated SCADA interface that enables real-time monitoring and control of all thermal subsystems. The SCADA platform provides a graphical overview of the full installation, representing major components, including the TCM reactor, heat pump, PCM tanks, solar field, inertia tank, and building load, using standardized symbols and interactive elements. Each subsystem is overlaid with live sensors and actuator data. Pale lime-green boxes display real-time values such as temperatures, pressures, flow rates, valve states, and pump statuses. These readings are retrieved from the PLC via Modbus TCP and updated continuously, ensuring synchronization with the physical system.

In Figure 11, a representative example of the SCADA interface during charging mode is shown. A control status panel located in the upper left section of the interface shows the global operating state (manual, automatic, or semi-automatic), TCM and PCM state of charge, and any active thermal demand requests. Below this, a user panel displays the seasonal mode (heating/cooling) and currently requested thermal services. An alarm module in the lower-left corner logs hardware and communication faults, offering immediate visual alerts. In the specific example shown in Figure 11, the red-highlighted loop corresponds to a charging state in which heat from the solar field and the TCM reactor is routed through the inertia tank and the heat pump to charge both hot PCM tanks (space heating and DHW) while simultaneously supplying the fan coil in the demo room, so that the active energy flow path can be identified at a glance by following the red pipelines.

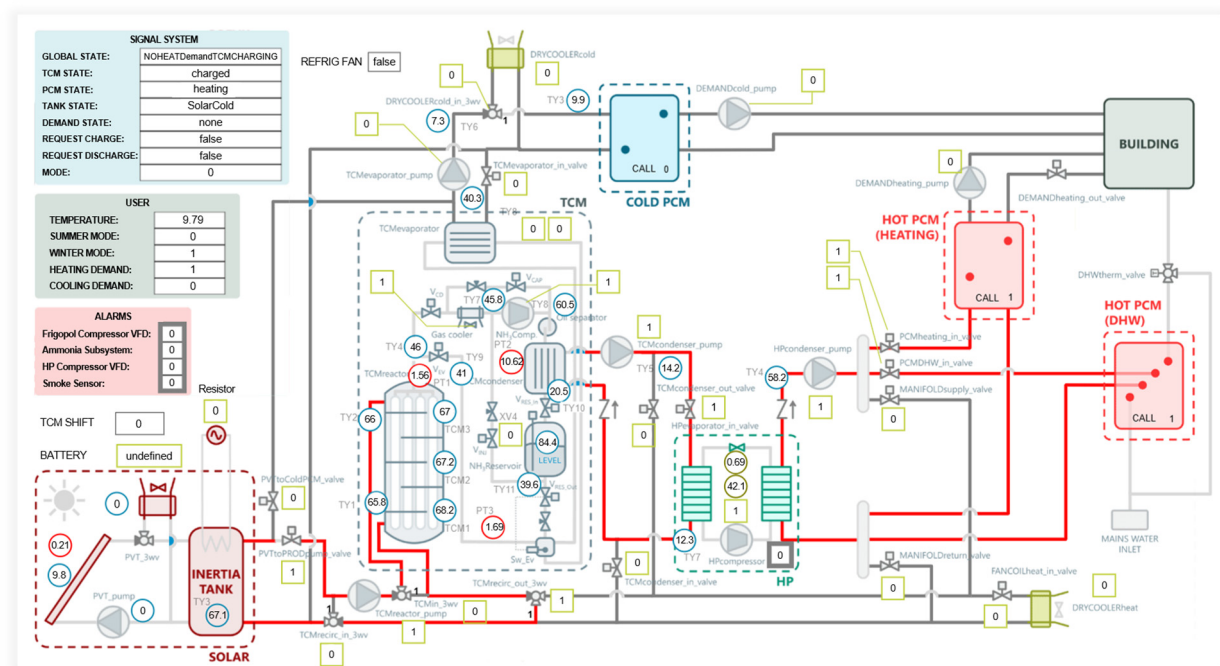


Figure 11. Example of use of the SCADA system. Charging TCM and PCMs.

Thermal pipelines are dynamically color-coded: grey when inactive, red when a flow path is active. This feature provides immediate visual correlation between the activation of valves and pumps and the resulting energy flows. SCADA supports three hierarchical control modes:

Manual Mode: Each actuator (pump, valve) can be controlled individually through on-screen toggles. This mode is essential for commissioning and testing.

Automatic Mode: A centralized EMS logic takes full control of system operation based on internal rules and environmental inputs.

Semi-Automatic Mode: The system recommends operational actions based on logic and waits for operator confirmation, allowing guided operation with human supervision.

It is important to note that the objective of the proposed EMS is not to globally optimize energy flows or to perform long-term planning under uncertainty, but rather to guarantee safe, stable, and explainable operation of a highly integrated hybrid storage prototype. As such, the control strategy prioritizes operational feasibility, component protection, and reproducibility over optimality. More advanced optimization-based or predictive EMS formulations are therefore intentionally excluded at this stage and are

considered a natural extension for future work once large-scale deployment and grid-interactive operation are targeted.

2.4. Monitoring and IoT

The system was structured around Node-RED, which is used as a field data-collection environment responsible for acquiring sensor signals and internal system states via industrial protocols such as Modbus and MQTT. The data collected by Node-RED were published in real time via the MQTT protocol to a centralized Mosquitto-based broker (Figure 12).

Once in the broker, the data were consumed by the Telegraf plugin, which acted as a collector and preprocessor, filtering, structuring, and forwarding the information to an InfluxDB database, where it was stored in time-series format. This database formed the core of the subsequent monitoring system.

This entire data acquisition and transmission methodology is complemented by a fault-tolerance strategy designed to prevent data loss in the event of temporary unavailability of the InfluxDB database service.

Finally, the Grafana platform (Figure 13) was used as the main visualization interface for analyzing and monitoring the operational data collected. Grafana, directly connected to the InfluxDB database, enables both real-time and historical representation of the system's most relevant variables, facilitating diagnosis, validation, and optimization of thermal and electrical cycles.



Figure 12. Data-sending methodology diagram.

Figure 13 presents a representative Grafana dashboard used during commissioning and normal operation of the MiniStor prototype. In this example, in the upper panel, the time series “Demo_outdoor” shows the outdoor air temperature at the demo site, while “HTF_TY_3” and “NH3_TY_1” correspond respectively to the inertia tank temperature and the TCM reactor temperature. The “PT100” trace represents the hot PCM temperature and “NH3_PT_1” the reactor pressure. The discrete signal “MODE_TCM_UNIT” indicates the active operating mode of the TCM unit (values 1–2 for charging and 3–4 for discharging), allowing events in the thermal and pressure curves to be directly linked to specific modes. In the lower panel, “Deltasol_Temperature_sensor_1” and “Deltasol_Temperature_sensor_2” correspond to the PVT outlet and solar collector inlet temperatures, while “Irradiance_WStation” reports the global irradiance from the on-site weather station. Rather than providing a detailed performance analysis, this figure illustrates the level of detail and customization of the Grafana dashboards, which enable operators to correlate solar resource, system operating mode, and key thermal variables in real time.

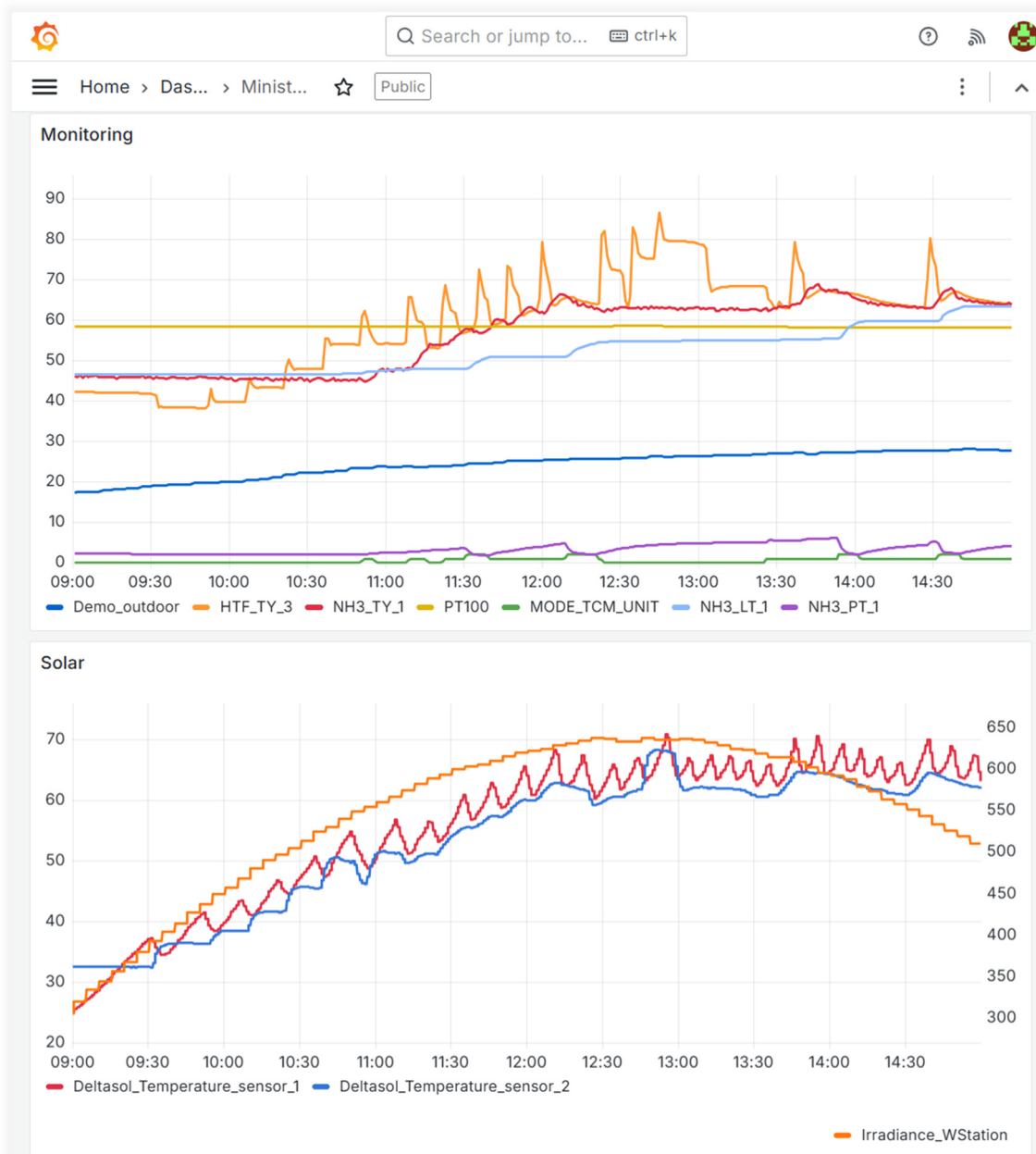


Figure 13. Grafana dashboard.

3. Data Management

Data acquisition, storage, and monitoring were implemented through a centralized Internet of Things (IoT) data management platform following a hub-and-spoke architecture. All on-site devices transmitted data to local gateways, which accumulated and forwarded the measurements to the central platform through a secure representational state transfer application programming interface (REST API) using JavaScript object notation (JSON) data structures. The platform frontend has been implemented in Angular [22], and the backend in Node.js [23], whereas two dedicated databases have been used: MongoDB for device, user, and the site metadata, and InfluxDB for time-series monitoring data.

The data monitoring process involves the propagation of data from the data transmission devices of each demo site, after accumulating all the necessary data locally on gateways, to the respective databases through the REST API (Figure 14). Then, the UI of the IoT platform retrieves the data to create intuitive plots and optimize the operation of

the system. The API handles the data in JSON format, which is a standard and human-readable file format that is generally used for server communication. The JSON format consists of key-value pairs or key-serializable-value pairs for more complex structures.

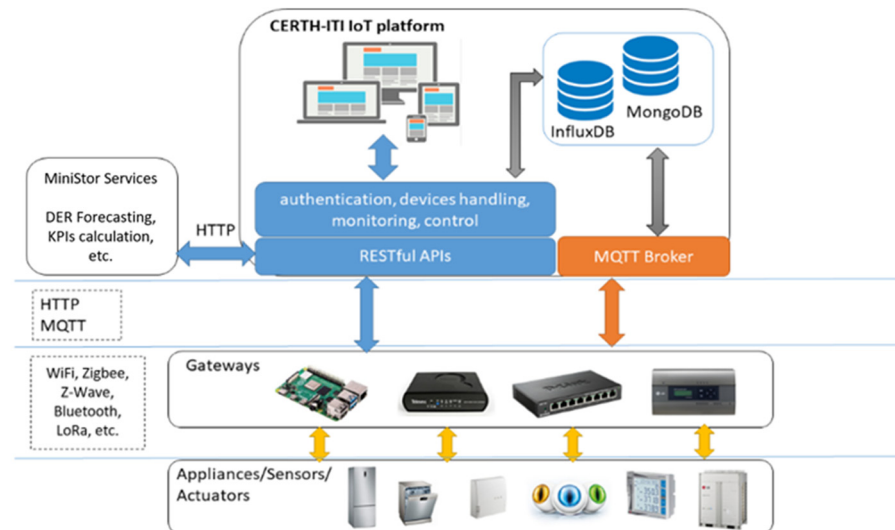


Figure 14. IoT MiniStor platform data.

Security is ensured by following approaches that comply with the core information security principles of availability, integrity, and confidentiality [24]. All data transfers occurred over hypertext transfer protocol secure (HTTPS), and access to the REST API required user authentication and role-based permissions. The API controlled all data ingestion and retrieval processes, ensuring traceability, controlled access, and anonymization where necessary. These properties are achieved by the utilization of a secure REST API that allows access to the data through encrypted user authentication processes, which also promotes data anonymity through controlled access.

Processed and raw data can be accessed either programmatically through authenticated REST API endpoints or via the platform's graphical user interface, which supports monitoring, visualization, and data-analysis tasks. This method ensures consistent data handling from acquisition to long-term storage, reproducibility of the monitoring workflow, and secure user-level access to system performance information.

The API offers multiple endpoints, but the main ones are the site, user, tenant, assignments, device, specification, and asset (Figure 15). The asset and specification endpoints are used for the descriptive properties of the devices, while the device endpoint is used to assign a token to a device that is used as device id for monitoring data endpoint (assignments). Finally, the assignments endpoint provides the upload and fetch functions for monitoring data.

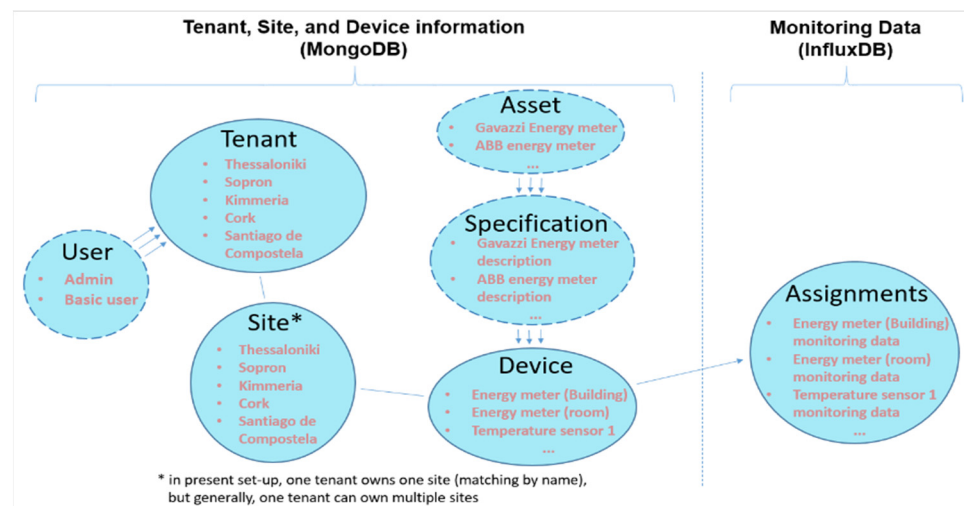


Figure 15. Main endpoints of the IoT's MiniStor platform REST API.

All the information is provided through a user-friendly interface (Figure 16), where the user can see a summary of the systems and environment's condition, including both internal and external conditions. External conditions are based on information from the local weather station. On this tab, the user can monitor the dwelling's energy consumption, find information about the status of the battery, its stored energy level, and whether it is charging or giving energy to the building. Finally, they can see what part of the energy needs of the dwelling are covered by the MiniStor system as well as the amount of energy that is exported or imported from the grid on a daily basis. There is also the data analytics tab, where the user can see the electrical and thermal production and consumption of the solar array as well as the correlated costs from the system's usage in the form of graphs.

Data were managed following a structured data management plan (DMP) aligned with FAIR (findable, accessible, interoperable, and reusable) principles. All raw and processed data were stored on an institutional secure server with encrypted daily back-ups. Version control was maintained using Git for analysis scripts and data preprocessing steps. Data cleaning followed predefined standard operating procedures (SOPs), including automated validation checks. Metadata and documentation accompanied each dataset to ensure reproducibility.

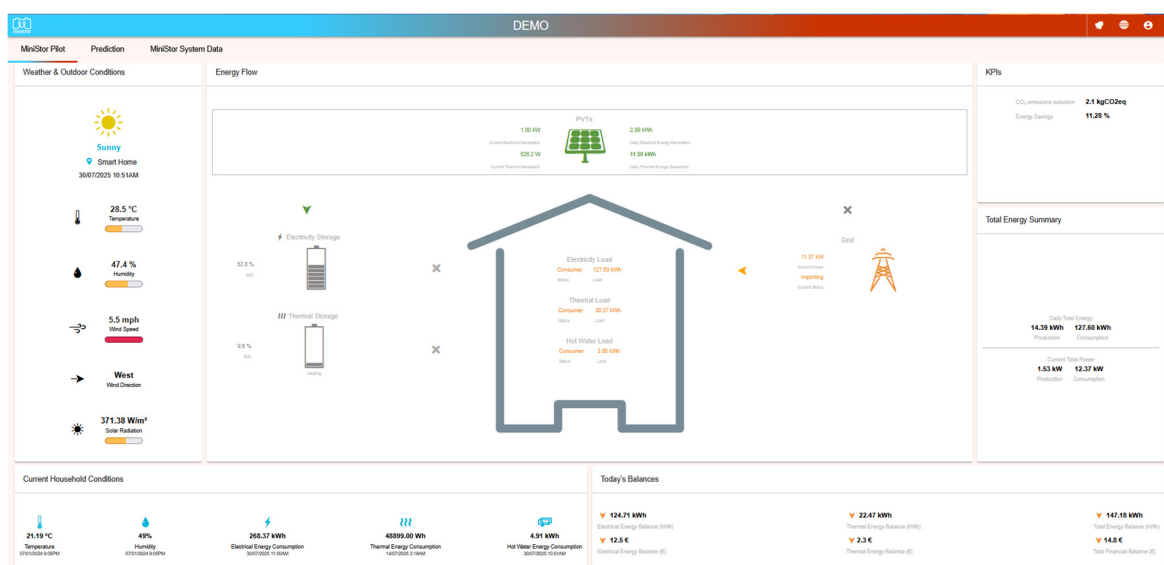


Figure 16. UI of the IoT MiniStor platform—Dashboard.

4. System Operation

This section presents the operation, including the mode transition and component-level control of the MiniStor TCM subsystem, the auxiliary heat pump (HP), and the hot/cold PCM buffers during a full charging and discharging cycle, utilizing the data from the monitoring and data management platforms as previously discussed. The high-level controller running on the IoT platform issues charge and discharge requests and temperature setpoints based on building demand and solar availability, while the low-level PLC and Node-RED logic implement the rule-based finite-state machine described in Section 2.3 to execute the corresponding operating modes and enforce all safety constraints. The supervisory logic of the high-level control collaborates with the low-level control unit for each TCM operating mode—pressurization/pre-heat (Mode 1), charging (Mode 2), depressurization (Mode 3), and discharging (Mode 4)—including their entry/exit criteria, safety measures, and the role of the solar/HTF loop. The cross-subsystem coordination is discussed in terms of safety measures and representative time-series behavior using Figures 17–24.

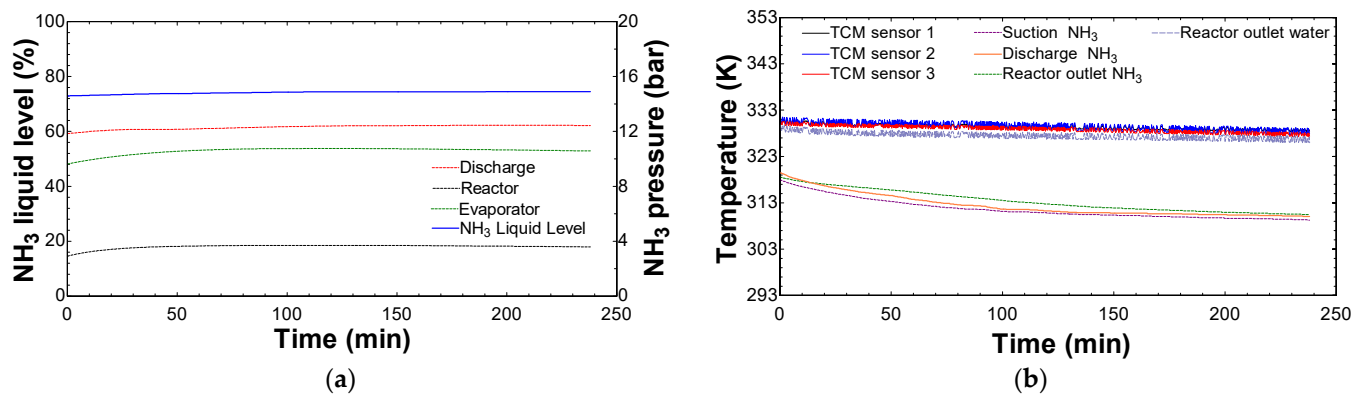


Figure 17. Standby mode: (a) liquid level at the receiver and pressure variation; (b) temperature variation within the TCM during standby.

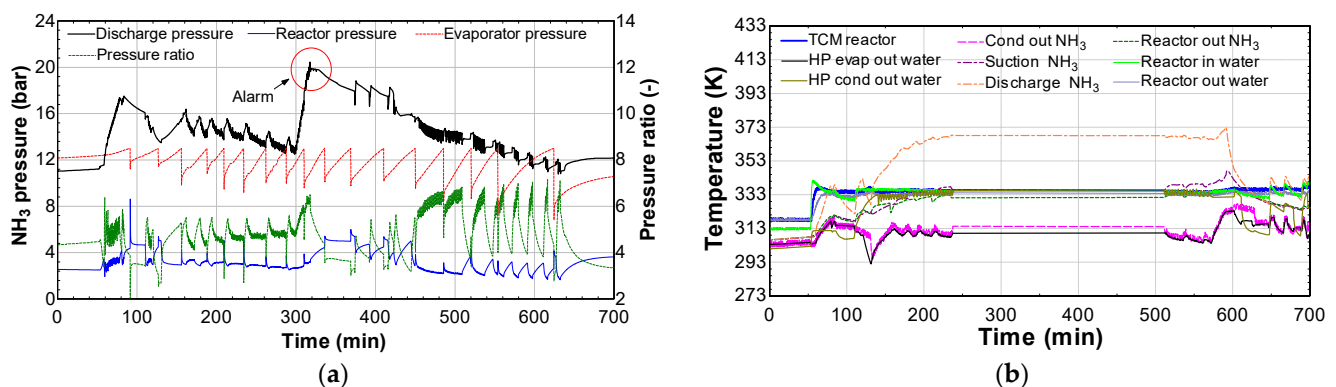


Figure 18. (a) Pressure variation of the TCM subsystem during charging mode; (b) temperature variation in the TCM subsystem during charging mode.

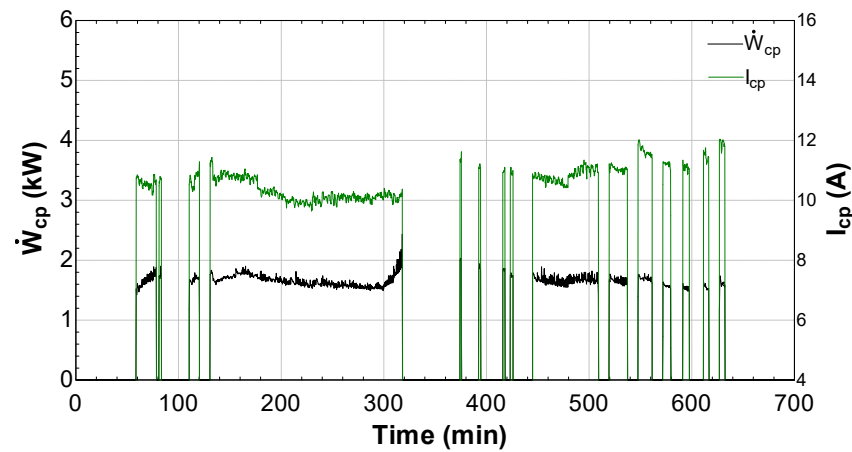


Figure 19. Electrical consumption of the NH₃ compressor during the charging phase.

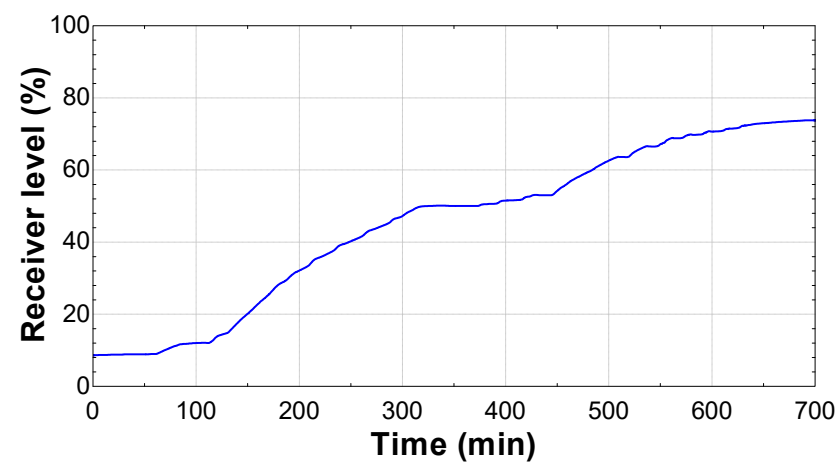


Figure 20. Time evolution of the NH₃ receiver liquid level during the charging phase.

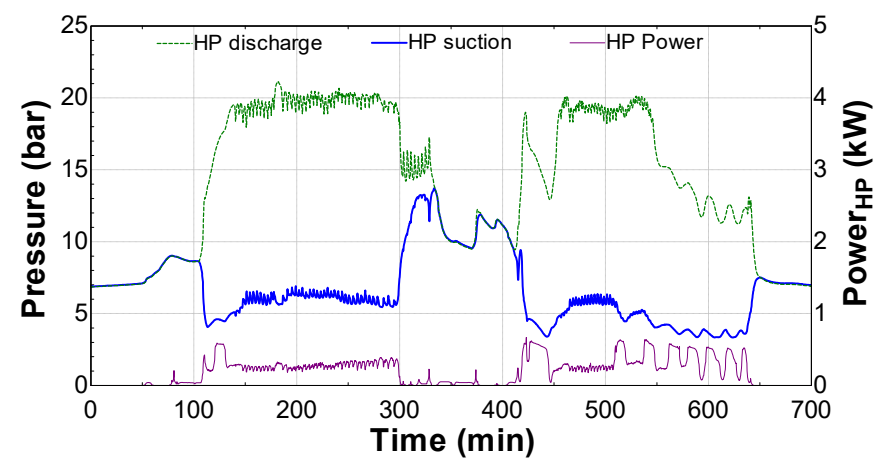


Figure 21. Electricity consumption and refrigerant pressure of the VC heat pump during charging mode.

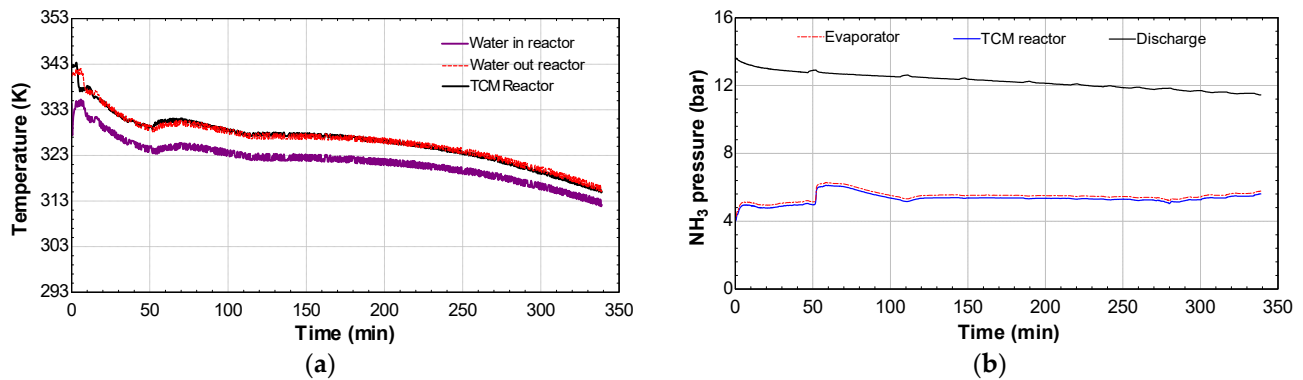


Figure 22. (a) Reactor temperatures at the NH₃ and water side during discharging mode. (b) NH₃ pressure levels of the TCM subsystem during discharging mode.

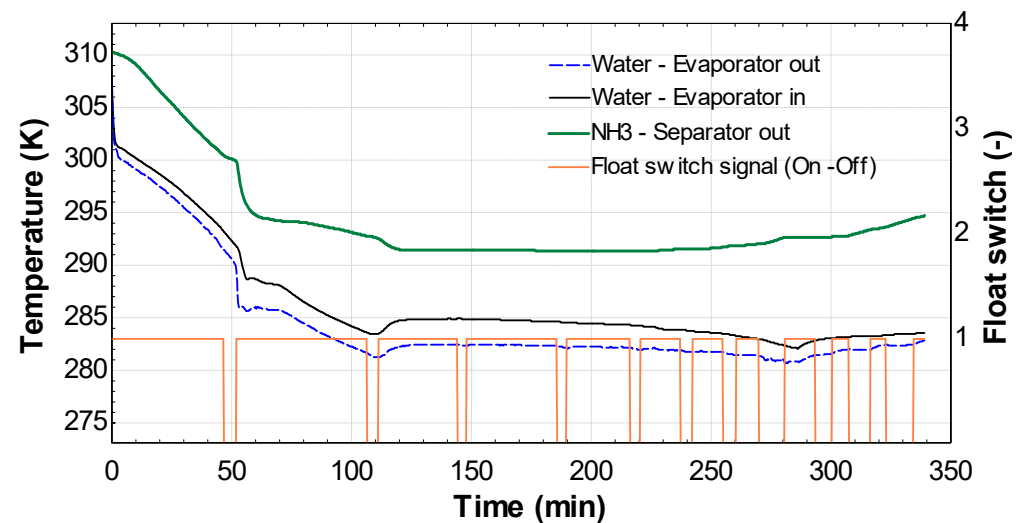


Figure 23. NH₃ and water temperatures at the evaporator of the TCM subsystem and float switch operation of the NH₃ separator during discharging.

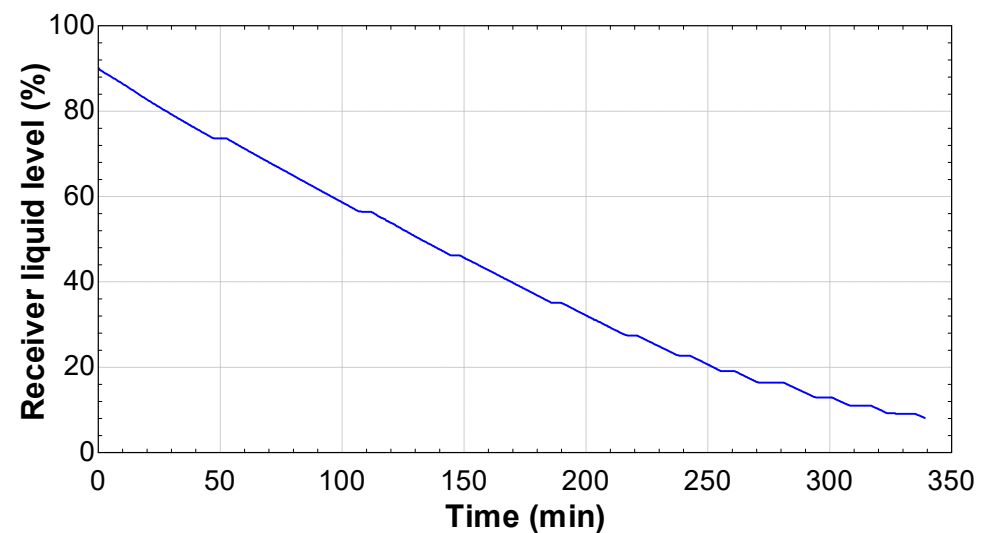


Figure 24. Time evolution of the NH₃ receiver liquid level during the discharging phase.

4.1. Standby

Thermal service is requested but not activated when suitable heat sources are unavailable (e.g., insufficient solar temperature), when no charge/discharge command has been issued, or when a safety interlock is active. All process valves are closed; pumps

are off except for periodic anti-seize jogs. The compressor VFD is inhibited, and the NH_3 receiver and reactor are hydraulically isolated. The low-level PLC continues to poll sensors, and the high-level control maintains the communication and performs alarm management. During standby the only operation that can be enabled is the defrost cycle, which is activated if any temperature falls below 274 K. During defrost, the circulating pumps and the solenoid valves in the heat-removal grid are activated circulating the heat-removal fluid over the whole network and keeping the temperature within acceptable limits.

Figure 17a shows the evolution of the three NH_3 pressures (discharge, reactor, evaporator) together with the receiver liquid level over a ~250 min window during standby mode. The NH_3 liquid level remains essentially flat indicating that the receiver is maintained properly isolated, and no significant net transfer or boil-off occurs during this period. The pressures exhibit small, smooth drifts due to the temperature variation in NH_3 with a persistent separation between evaporator, reactor, and discharge, showing values of approximately 10, 12, and 4 bar, respectively.

Figure 17b shows the temperatures for the TCM reactor (TCM sensors 1–3), gas lines (NH_3 suction/discharge and reactor-outlet NH_3), and reactor-outlet water. TCM temperatures are quasi-stable around the 328–333 K range with only a slight downward drift, due to minor heat losses. Suction and reactor-outlet temperatures decrease mildly over time, in the range between 308 and 318 K due to the thermal losses at the piping network.

4.2. Pressurization and Charging Phase

Prior to the initiation of the TCM charging phase, the pressurization sequence (Mode 1) from the standby mode is required. With the reactor isolated and all NH_3 side valves closed, the HTF pump is enabled once the solar buffer exceeds a temperature setpoint (usually > 328 K), and hot HTF circulates through the reactor jacket while the NH_3 circuit remains locked. As jacket heating proceeds, the reactor temperature rises to ~339 K and the reactor pressure increases rapidly toward ~2.5 bar, consistent with the Clausius–Clapeyron dependence of NH_3 vapor pressure on temperature. This pre-heating process causes the reactor material to desorb vapor NH_3 , elevating the suction pressure of the compressor. Throughout most of Mode 1, the receiver liquid level remains essentially constant—no mass transfer is expected while the valves are closed and the compressor is idle. According to the TCM control specification, Mode 1 terminates when the tank pressure exceeds its setpoint (~2.5 bar) and is re-entered if the measured compressor pressure ratio later surpasses its limit (e.g., $\text{PR} > 5.5$) or the TCM pressure falls below the threshold of 1 bar.

Continuing with the charging phase, Figure 18a,b captures the supervisory logic that alternates between Mode 2 (charging— NH_3 compressor active) and Mode 1 (pressurization) to keep the TCM subsystem within its envelope, ensuring safe operation. During charging mode (from min 100 to min 600), as shown in Figure 18a, the boundary conditions imposed on the reactor were kept within a narrow band, showing an average value of 329.5 K at the inlet temperature to the reactor and typically ranging ± 1.5 –4 K, while the HTF mass flow rate was kept constant (about $0.4 \text{ kg}\cdot\text{s}^{-1}$) through a variable speed pump operating at constant flow mode. These measures ensure that externally induced fluctuations did not propagate into the TCM system and affect the control logic. Consequently, all transitions between Mode 2 and Mode 1 during the analyzed window were solely triggered by the internal process variables of the NH_3 –TCM circuit, and specifically by the evolution of compressor pressure ratio, the suction and discharge pressures and temperatures, and the condenser-side temperature conditions linked to heat

pump stabilization. All these rule-based logic controls are discussed in the following paragraphs.

In Figure 18a, the evaporator-side pressure exhibits a small periodic modulation during charging, due to local temperature fluctuations at the pressure sensor. In Mode 2 this part is expected to be stable because that branch is isolated from the rest of the TCM network. For the analysis, a time window was selected that includes both long periods of normal operation and incidents where protective actions are invoked. Mode selection is performed by the low-level control based on rule-based techniques. To enter Mode 2, the TCM reactor pressure must exceed 3 bar. The thermal power absorbed by the TCM reactor is related to the ability of the compressor to compress a given mass flow rate of NH_3 at a given rotation speed and compression ratio. It also depends on the ability of the reactor to absorb the thermal power provided by the solar array, in particular the quality of the heat exchanged between the HTF and the wall of the reactor and the wall with the reactive medium. As a result, thermal power supplied to the TCM reactor causes the desorption of NH_3 from the reactive salt, enabling the TCM reactor to produce an NH_3 mass flow rate ranging from 2 to 7 kg/h as the power increases from 1 to 5 kW. The NH_3 compressor (Frigopol 7DLY) is driven by a dedicated VFD, allowing motor frequency adjustment from 30 to 50 Hz—i.e., a shaft speed of 1800–3000 rpm. Over this range, the compressor's swept volumetric flow varies proportionally from approximately 3.5 to 7.0 $\text{m}^3 \text{h}^{-1}$, enabling precise matching of desorption throughput to operating conditions. Conversely, if the reactor pressure were to fall below 1 bar while the compressor is running, charging is aborted, and the system reverts to Mode 1. In the time series shown, this lower bound is never reached, achieving adequate suction pressure for the whole charging period.

A second and more stringent criterion is the compressor pressure ratio, which must remain below 6.5 to fall inside the compressor operating map. If pressure ratio above 6.5 persists for 1–2 min, charging is stopped, and Mode 1 is re-entered. This behavior is visible after 500 min in Figure 18a, where gradual depletion of the reactor driving force lowers suction pressure. Despite the compressor operating already at its minimum speed (30 Hz), pressure rises and triggers a protective transition. In normal operation, a target pressure ratio of 4 is imposed with a PID regulator adjusting compressor speed to track this target, reducing frequency down to 30 Hz when PR climbs, thereby easing the condenser pinch point. Moreover, two additional temperature safety measures protect both the NH_3 compressor and the vapor-compression (VC) heat pump that upgrades desorption heat for storage in the PCM tanks. The HTF inlet to the NH_3 condenser must remain $<313 \text{ K}$, and the water temperature at the condenser heat exchanger must remain $<323 \text{ K}$. If either limit is exceeded during charging, the compressor is stopped, and Mode 1 is enabled to allow the VC heat pump to pull the condenser side back to acceptable levels. This start-up interaction is evident between 70 and 130 min. In this period the rapid increase in suction pressure due to an initially fast TCM reaction elevates discharge pressure and condenser-side water temperature before the VC heat pump reaches its operating point, prompting brief charging interruptions. After ~130 min the heat pump reaches steady operation, and the system sustains continuous charging for ~3.2 h (130–320 min) with all constraints respected, the compressor pressure ratio remains within limits, and all temperature safety measures are satisfied. Throughout this interval the pressure ratio is initially elevated and the VFD holds the compressor near its minimum speed (30 Hz). As desorption progresses, the reactor pressure rises, increasing suction pressure and thereby reducing pressure ratio, the controller then commands a higher speed to track the PR setpoint, which in turn moderates the pressure rise.

A final noteworthy event in Figure 18 illustrates the low-level protective response when operating variables approach their acceptable limits. At 320 min, the desorption

rate accelerates, driving the TCM reactor pressure to ~5 bar. Concurrently, the compressor pressure ratio rises above 5, forcing the VFD to the minimum speed of 30 Hz. Despite this, the discharge pressure continues to climb and reaches the 20 bar trip threshold, which triggers the high-pressure alarm, terminates Mode 2 (charging), and places the system in standby until pressure is reduced. The subsequent restart sequence resumes Mode 1 (pressurization), but because suction pressure remains elevated, the pre-heat criteria are immediately satisfied, and the system returns to Mode 2 after the prescribed delay. A second stabilization phase follows between 370 and 430 min, during which the controller alternates between Mode 1 and Mode 2 in compliance with the limits of HTF inlet temperature to the NH₃ condenser. After the high-pressure trip, the condenser-side temperature remains temporarily high and the heat pump has not yet managed the heat produced by the TCM subsystem, so the controller enforces short duty-cycle interruptions of charging until the condenser inlet is brought back within the <313 K limit.

Upon entering pressurization and charging mode, the reactor jacket brings the TCM reactor temperature to approximately 338 K, a few degrees below the HTF inlet to the reactor, as expected. Later in the record ($t > 450$ min), the higher solar-field supply increases the reactor temperature further, approaching 343 K.

The NH₃ suction temperature rises progressively from its pre-heat value of 306 K to 341 K, tracking the gradual warm-up of the TCM loop under compressor operation. A similar upward trend appears on the discharge side, with a ~30 K ΔT between discharge and suction temperature of the compressor during stable operation. The maximum discharge temperature of ~373 K is observed with the high-pressure event when discharge pressure briefly reached 20 bars. On the high-pressure side, the condenser-out NH₃ temperature closely follows the HP evaporator water outlet, revealing an effective heat transfer between the TCM condenser and HP evaporator plate heat exchanger. The modest oscillations visible on these traces reflect the heat pump capacity modulation and variations on the load side. Downstream, the HP condenser-out water remains within 328–338 K, confirming reliable heat upgrading and compatible supply conditions for charging the hot PCM storage tank.

In Figure 19, in agreement with the previous remarks, the NH₃ compressor electrical power consumption and line current are shown over the same time window as in Figure 18a,b. During the stabilized charging interval (~130–320 min), power remains in the 1.8–2.3 kW band while current ranges between 9 and 11 A, consistent with the variation in the operating pressure ratio conditions and almost always within 30 Hz rotational frequency. Transient current spikes visible at the starting boundaries of charging phase correspond to start events commanded by the low-level regulation between Mode 1 and Mode 2. Additionally, the intervening gap (~320–420 min) shows the period when the effect of high-pressure alarm occurred, and then, repeated dropouts to zero power/current, indicate protective transitions. In the second charging block (~420–610 min), power returns but with slightly lower mean and more frequent cycling, since the TCM reaction speed is reduced at the end of the process.

In Figure 20, the compressor operation and charging process is reflected to the liquid NH₃ level of the TCM subsystem receiver. At the initial ~0–300 min, after a brief period (<100 min) when standby mode is active, the level rises with a steep, nearly linear slope, indicating that the reactor provides ample NH₃ vapor and the compressor operates continuously, corresponding to favorable kinetics, strong thermal driving from the jacket, and stable system operation. In the mid-cycle period, ~300–420 min, the curve flattens, reflecting the low-level control interruptions which temporary stops the charging process leading to zero net mass flow at the NH₃ receiver. During the late-cycle period, ~420–700

min the charging resumes but the increasing slope progressively decreases as the reaction driving force weakens and suction pressure drifts downward.

Finally, in Figure 21 the suction and discharge pressures of the VC HP together with the HP compressor electrical power is depicted. Three operating phases are evident and map directly to the charging process discussed earlier. During the start-up and first continuous charging block (~70–320 min) the HP operates in a steady mode with discharge pressure ranging between 18 and 20 bar and suction pressure in the 5–7 bar band, consistent with lifting desorption heat from the TCM condenser to a 328–338 K water supply. The electricity consumption of the compressor is calculated from the polynomials available from the manufacturer according to the pressure ratio and rotational speed and for this operating phase and has an average value of 350 W, with small variations reflecting capacity modulation. The relatively constant discharge and suction levels indicate stable evaporator/condenser boundary conditions at the plate heat exchangers. During the protective interruption and recovery (~320–420 min), the TCM compressor due to the high-pressure event halts its operation, which propagates to the HP lower discharge pressures. The controller responds by suspending charging and the HP after a time delay compressor power drops to ~0 kW.

In the third phase, at the late-cycle period (~420–610 min) the charging resumes along with the HP operation, discharge pressure again stabilizes initially near 18–20 bars, then gradually declines toward ~12–14 bars as the load and available TCM desorption heat and condensing temperature decrease. Suction pressure follows the same trend, falling from ~6 bar to ~3–4 bar, as the reactor driving force weakens. Additionally, the increase in the operating pressure ratio along with the compressor speed regulation leads to higher electricity consumption, up to 0.8 kW.

4.3. Depressurization and Discharging Mode

When cold is requested and stored liquid NH_3 in the receiver is available, the supervisory controller (high-level and low-level control) commands Mode 3 (depressurization) and then Mode 4 (discharging). In this phase, neither the NH_3 compressor nor the VC heat pump operate, making this process simpler to control. In Mode 3, all NH_3 -side valves remain closed, and the reactor is hydraulically isolated. Since the presented data from discharging phase concern a summer-period operation where only cooling is required, the HTF circuit is routed through a heat-rejection path (fan-coil to ambient or, if beneficial, the hot-PCM loop operated as a sink) and circulated through the reactor jacket. The objective is to lower the reactor temperature and pressure to a temperature target (<333 K), so that the reactor pressure is lower than the evaporator pressure by a small, controlled margin. This condition ensures a stable vapor driving force from the evaporator toward the reactor once the discharge begins.

Upon satisfaction of this inequality, the controller transitions to Mode 4 (discharging). Liquid NH_3 from the receiver is transferred by gravity to a flooded evaporator, whose level is maintained by an on/off float switch located in a small liquid–gas separator placed upstream. The evaporator extracts heat from the cold-PCM circuit, while the produced vapor flows freely back to the reactor where it is absorbed by the salt, causing an exothermic reaction whose heat is rejected to ambient via the HTF loop. Discharging continues until the receiver liquid level reaches the lower liquid level threshold (e.g., 8%), at which point the mode is terminated.

Figure 22a shows a discharging period focused on cooling production. Because the reactor's absorption heat is rejected to ambient during this season, the reactor temperature exhibits a monotonic decline from approximately 338 to 317 K, even though absorption is exothermic. The mean jacket temperature lift (reactor outlet minus inlet) is about 5 K, consistent with steady removal of the absorption heat at modest HTF flow. A brief

increase around 50–70 min coincides with a change at the hydraulic configuration in the HTF circuit with warmer fluid entering the reactor, increasing slightly the TCM temperature and after that the cooling trend resumes and the reactor continues to unload heat to the ambient loop.

Figure 22b presents the corresponding NH_3 pressures in the evaporator, reactor, and the condenser side. The condenser branch is inactive and isolated throughout Mode 4; hence, its pressure remains essentially constant apart from small ambient-temperature variations. In contrast, the reactor and evaporator pressures track each other closely, with the evaporator held ~ 0.1 – 0.2 bar higher than the reactor, due to the depressurization phase, to sustain vapor flow from the evaporator back to the reactor. Both pressures exhibit similar co-variation with the reactor temperature in Figure 22a, reflecting the free communication between these two parts in the TCM subsystem. Additionally, the pressure levels for most of the discharging time remain nearly constant, implying a quasi-constant evaporating temperature in the flooded evaporator and, therefore, a stable cold-production condition for charging the cold PCM.

During cold delivery to the PCM the evaporator circuit simultaneously redirects the chilled NH_3 flow toward the cold-PCM loop, lowering its temperature until the PCM reaches its phase-change setpoint of 11°C .

In Figure 23, the response of the flooded NH_3 evaporator and its level control during Mode 4 is shown. At the onset of discharging, the secondary (water) loop enters the evaporator at about 301–303 K and exits with a 1 K lower temperature. As the process advances, both inlet and outlet temperatures decrease monotonically, while the liquid NH_3 absorbs heat from the HTF. A short discontinuity around 50–70 min in accordance with the previous remarks corresponds to a briefly introduced warmer fluid, which potentially increases the absorption speed of the reaction, increasing with this way the available cooling power showing a sharp decrease in the HTF temperature at the evaporator side. Thereafter, the process returns to a smooth decline, and the progressive narrowing of the inlet–outlet gap after ~ 250 min reflects the reducing load as the PCM nears its limit.

The refrigerant side NH_3 temperature at the separator outlet exhibits a rapid drop during the first hour followed by a narrow, slowly varying band for the remainder of the record. This behavior is consistent with a quasi-constant evaporating condition in the flooded evaporator and indicates that the two-phase mixture is being maintained without any dry-out. The absence of pronounced oscillations demonstrates that the separator mixing volume effectively damps short-period disturbances originating from the liquid-level actuation and load fluctuations.

On the other hand, the liquid-level regulation in the separator is provided by a float switch which commands the liquid-supply valve, placed at the outlet of the liquid receiver. When the level of the float switch is not reached, the sensor's indication is 1 (ON) as a normal closed contact, and when the NH_3 liquid has filled the sensor's position the signal returns to 0 (OFF). The switching trace, as shown in Figure 23, is well spaced discrete pulses early in the discharge, when the boiling rate is higher and the separator operates for longer periods. When the discharging process continues, the separator holds more liquid and progressively tighter spacing as the water temperatures approach 283 K and the thermal duty declines.

Lastly, Figure 24 summarizes the discharging sequence by tracking the receiver liquid level as a function of time. The curve exhibits a smooth, monotonic decline from a high initial inventory to the terminating threshold (around 5%), with two features consistent with the thermo-hydraulic behavior discussed above. First, during the early stage the slope is approximately linear, reflecting steady flooded evaporation and uninterrupted vapor return to the reactor, conditions confirmed by the stable refrigerant tem-

perature at the separator outlet and the well-behaved long cycles of float switch on–off regulation. Second, as the process advances due to the gradual reduction in the absorption driving force, the decreasing water-side temperature as the cold PCM approaches its phase-change setpoint, and due to the supervisory pauses introduced by the level controller’s hysteresis, a progressively slight decrease in the level slope is observed. Additionally, the absence of upward steps indicates no back-flow or flashing toward the receiver.

5. Conclusions

The MiniStor project has demonstrated the successful design, development, and deployment of a novel compact hybrid thermochemical–PCM thermal energy storage (TES) system integrated with advanced power electronics for sustainable heating, cooling, and electricity storage in residential buildings. The system harnesses a high-performing $\text{CaCl}_2/\text{NH}_3$ thermochemical reaction combined with hot and cold PCM, delivering an exceptionally high overall energy storage density. This compact design effectively addresses the limitations of existing TES technologies by enabling long-term, seasonal thermal storage in a minimal footprint, making it highly suitable for retrofitting existing building stock in European cities. It further highlights the transformative potential of integrating advanced electronics with renewable energy and high-density thermal storage in residential buildings:

- **Electronics-enabled control and optimization:** Multi-level converters and VFD-driven compressors enable precise management of energy flows between the solar field, TCM reactor, heat pump, and PCM storage.
- **IoT and SCADA integration:** Real-time monitoring and supervisory control platforms facilitate predictive operation, fault detection, and automated response, demonstrating how electronics and communication technologies enhance system reliability.
- **Efficient thermal–electrical coordination:** The combination of TCM–PCM storage with intelligent power electronics supports stable heating and cooling delivery, demonstrating the feasibility of compact, high-density energy storage systems in smart buildings.
- **Scalability and adaptability:** Modular electronics and digital control architectures provide a blueprint for flexible deployment in urban energy networks, enhancing RES integration.

All features considered, MiniStor is a prime example of how sophisticated electronics, control schemes, and IoT-enabled monitoring may hasten the implementation of robust, efficient, and sustainable building energy systems. Overall, this work confirms that the synergy between emerging power electronics technologies and advanced thermal energy storage solutions is vital in accelerating the transition toward sustainable, efficient, and resilient energy systems for heating and cooling. Continued research and development in component materials, control algorithms, and system integration will be instrumental in overcoming the remaining technical barriers and achieving large-scale adoption. Future work will focus on extending the high-level EMS toward grid-interactive operation and more advanced optimization frameworks, potentially including two-stage robust formulations and the explicit use of building thermal inertia and external waste-heat sources as additional flexibility options, in line with recent developments in low-carbon operation of remote hybrid energy systems. Furthermore, recent developments in decentralized resilience-oriented restoration frameworks, such as those coordinating building thermal flexibility with external mobile resources, may also

offer inspiration for long-term research directions, although such mechanisms lie beyond the scope and objectives of the present prototype-oriented study.

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Abbreviations

The following abbreviations are used in this manuscript:

CERTH	Center for Research, Hellas
DHW	Domestic Hot Water
EEV	Electronic Expansion Valve
EU	European Union
FPC	Flat-Plate Collectors
HP	Heat Pump
HTF	Heat Transfer Fluid
HTTP	Hypertext Transfer Protocol
HVAC	Heating, Ventilation, and Air Conditioning
IoT	Internet of Things
JSON	JavaScript Object Notation
MQTT	Message Queuing Telemetry Transport
PCM	Phase-Change Material
PLC	Programmable Logic Controller
PR	Pressure Ratio
PVT	Photovoltaic–Thermal
RES	Renewable Energy Source
REST API	Representational State Transfer Application Programming Interface
SCADA	Supervisory Control and Data Acquisition

TCM	Thermochemical
TCP/IP	Transmission Control Protocol/Internet Protocol
TES	Thermal Energy Storage
UI	User Interface
VC	Vapor-Compression
VFD	Variable Frequency Drive

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